INCH-POUND

MIL-I-8276C(USAF)
15 August 1994
SUPERSEDING
MIL-I-8276B(USAF)
13 October 1987

MILITARY SPECIFICATION

INDICATOR, RATE OF FLOW, FUEL, SYNCHRO STYLE, TYPE MA-2

This specification is approved for use by the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a single remote-indicating fuel flow rate indicator, in a two inch case, designated Type MA-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-1672	Boxes, Shipping, Reusable With Cushioning

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-B-131	Barrier Materials, Watervaporproof, Greaseproof, Flexible
MIL-I-7057	Indicators, Synchro, Aircraft, General Specification For
MIL-T-8275	Transmitter, Rate Of Flow, Synchro Style, Type MA-1

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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STANDARDS

FEDERAL

FED-STD-101	Preservation, Packaging And Packing Materials: Test Procedures

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packaging Requirements
MS33585	Pointers, Dial, Standard Design Of Aircraft Instrument
MS33639	Case, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric Integral Mounting

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Air Force Drawing No. 8988994

(Copies of this drawing may be obtained from Oklahoma Air Logistics Center/ TICLA, Tinker AFB OK 73145-5990.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements specified in MIL-I-7057 are applicable as requirements of this specification. Additional requirements shall be as specified herein. In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.1.1 Indicator. The indicator shall operate on the synchro system specified in MIL-I-7057.

3.1.2 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 6.3).

3.2 Design and construction.

3.2.1 Calibration. Each indicator dial shall be graduated in accordance with figure 1. The calibration of fuel flow versus angular degrees shall be linear within $\pm$ 0° 15' of arc, except at the changeover plate.

3.2.1.1 Transmitter. Unless otherwise specified, the indicator shall be designed to indicate, at a remote station, the rate of flow in pounds per hour of aviation fuel when used in conjunction with a fuel flow transmitter in accordance with MIL-T-8275.

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3.2.2 Pointer. The pointer shall conform to MS33585-8 except that the length of the pointer shall be such that the tip will overlap from 0.333 to 0.667 the length of the shortest graduation.

3.2.3 Fluorescent-luminescent markings. The following markings shall be finished in fluorescent-luminescent material per Air Force Drawing No. 8988994. The dimensions of the markings shall be as follows:

MARKINGS	HEIGHT OR LENGTH INCH ± 0.016	WIDTH OF LINE OR GRADUATION INCH ± 0.005
Scale Numerals 0 through 6	0.156	0.031
Scale Numerals 8 through 24	0.094	0.020
Numbered 1,000-pound graduations 0 through 6	0.156	0.031
Numbered 1,000-pound graduations 6 through 24	0.156	0.020
Intermediate 1,000-pounds graduations	0.125	0.020
500-pound graduations	0.125	0.031
100-pound graduations	0.094	0.020
Lettering "PPH x 1000"	0.094	0.016
Lettering "FUEL FLOW"	0.125	0.020

3.2.3.1 Durable dull black. The marking "TYPE MA-2" shall be permanently and legibly marked on the dial in 0.047 inch letters. This marking and all other markings, not otherwise specified, shall be finished in durable dull black.

3.2.4 Case. The case shall be hermetically sealed and shall conform to MS33639. The dimensions "L" shall be 2.750 ± 0.250.

3.2.5 Dial diameter. The dial shall have a minimum diameter of 1.750 inches as measured at the outside ends of the scale graduations.

3.2.5.1 Electrical receptacle. The indicator electrical receptacle shall be in accordance with MS33678-145-2P.

3.3 Weight. The weight of the complete indicator shall not exceed 0.6 pound.

3.4 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.5 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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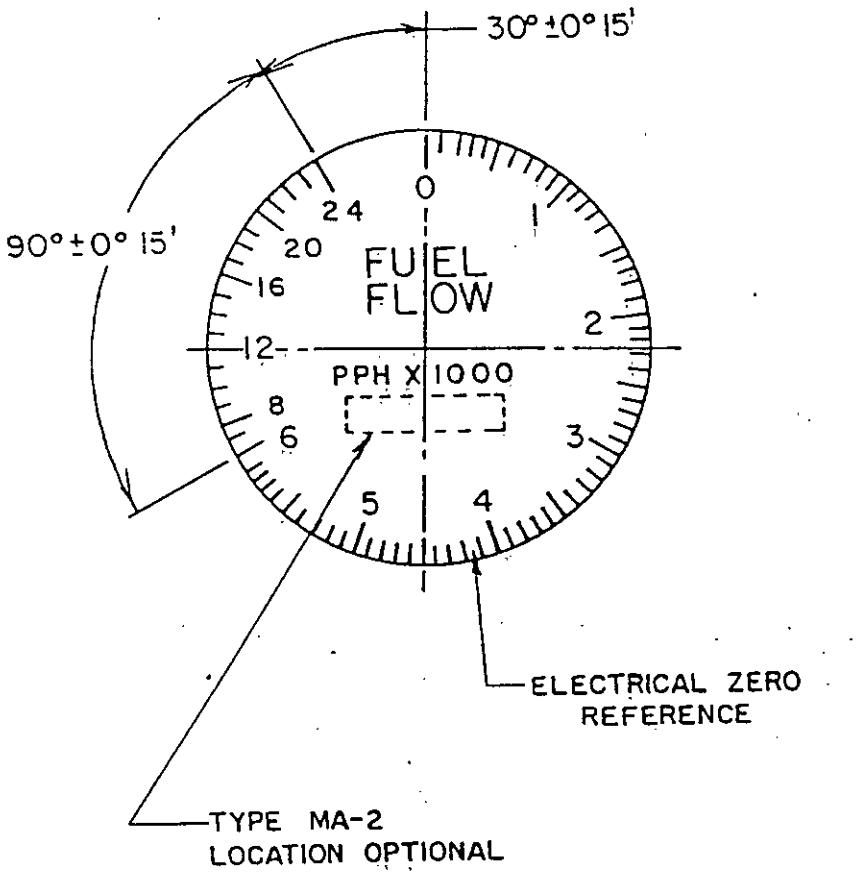


FIGURE 1. INDICATOR dial face

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General. The sampling, inspection, and test procedures shall be in accordance with MIL-I-7057.

4.2 Performance. The performance of the indicator shall be checked against Tables I and II of this specification.

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with section four of MIL-P-116 and the test conditions specified in this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level "A" or "C" as specified (see 6.2).

5.1.1 Preservation-packaging. Unless otherwise specified by the contracting activity, item whall be packaged in Quantity Unit Pack (QUP) of one each. Each item will be provided a preservation method IA-8 in accordance with MIL-P-116.

5.1.2 Cleaning. Item shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.3 Drying. Immediately after cleaning, the item shall be dried following any one or combination of the drying procedures listed in MIL-P-116. The drying procedures employed shall not be injurious to the item.

5.1.4 Level A. Item shall be preserved in accordance with MIL-P-116 and MIL-STD-2073-1 to provide a method IA-8.

5.1.5 Level C. Item shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity as specified in MIL-P-116 and MIL-STD-2073-1.

5.2 Packing. Packing shall be level "A," "B," or "C" as specified (see 6.2).

5.2.1 Container-cushioning. Unit container requirements shall conform to MIL-B-131, barrier material, unless otherwise specified by the contracting activity. Unit containers shall be large enough to allow for application of sufficient density and thickness to protect the unit container from punctures. Intermediate/hipping containers shall be large enough to allow for application of sufficient cushioning material, between the unit container and intermediate/shipping container. Cushioning material shall be of sufficient density and thickness to ensure shock transmission does not exceed peak values in G's established for the item.

5.2.2 Level A. Fiberboard containers do not meet Level "A" container criteria.

5.2.3 Level B. Item will be preserved as specified in 5.1 and shall be packed in exterior containers conforming to PPP-B-1672, weather-resistant unless otherwise specified by the contracting activity. Exterior container shall be uniform shape, size and minimum tare and cube, consistent with the protection required. Closuure shall be in accordance with appropriate PPP-B-1672 procedures and contracting activity.

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TABLE I. Indicator fuel flow rate tolerances

FLOW RATE PPH	Tolerance PPH ± ROOM TEMPERATURE	LOW TEMPERATURE	HIGH TEMPERATURE
400	20	30	20
1,000	20	30	20
2,000	20	30	20
3,000	20	30	20
4,000	20	30	20
5,000	20	30	20
6,000	20	30	20
8,000	200	300	200
12,000	200	300	200
16,000	200	300	200
20,000	200	300	200
24,000	200	300	200

TABLE II. Indicator test tolerances.

TEST	Tolerance PPH ± 0 - 6,000 SCALE	6,000 - 24,000 SCALE
Frequency and Voltage variation	20	200
Position Error at 5,000 PPH	20	---
Friction:		
Room Temperature	40	300
High Temperature	40	300
Low Temperature	65	500
Vibration:		
Pointer Oscillation	40	400
Pointer Variation	40	400
Endurance:		
Response Time Three Second Maximum	40	300

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5.2.4 Level C. Item will be packed in such a manner that will afford adequate protection against physical damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to MIL-STD-2073-1.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked in accordnace with special markings required by the contracting activity, and MIL-STD-129.

5.4 Inspection and test. Method of preservation tests shall be accomplished in accordance with section 4 of MIL-P-116 to ensure compliance with section 5 of this specification. Packaging tests shall be conducted in accordance with rough handling as specified in FED-STD-101.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Type MA-2 indicator covered by this specification is intended for use to indicate the rate of flow of fuel in pounds per hour consumed by an aircraft turbo-jet engine.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODIIS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Levels of preservation, packaging, and packing required.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No.) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Logistics Center/TICLA, Tinker AFB OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key words) listing. Recommended key words are:

Fuel Flow Rate Indicator
Synchro Style
Two Inch Case
Type MA-2

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-8276C(USAF)	941004/15

3. DOCUMENT TITLE
Indicator, Rate Of Flow, Fuel, Synchro Style, Type MA-2

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
OC-ALC/TICIA		
c. ADDRESS (Include Zip Code) 3001 Staff Drive TAFB, OK 73145-5990	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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